

Felipe Espinosa

Applied AI Engineer | NLP and LLM Systems

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EXPERIENCE

AI Evaluation & Data Contractor | Remote

Independent Contractor

November 2025 – April 2026

- Delivered coding, data scraping, and AI-data tasks for client projects across contractor platforms, using Apify and LLM-based assistants to collect, structure, and validate data.
- Evaluated LLM responses against strict rubrics for factuality, correctness, reasoning quality, and safety, including health-related model-comparison tasks.
- Analyzed failure patterns and prompt behavior, including unsupported claims, missing evidence, hallucinated facts, and weak reasoning, to support model-quality review.

Capital Fund Management (CFM) | Paris, France

AI Engineer Intern — LLM Systems

March 2025 – September 2025

- Built a conversational SQL agent for financial data analysis, achieving 87.5% end-to-end accuracy on a 48-question internal benchmark.
- Implemented a RAG retrieval pipeline using BM25, dense embeddings, and Reciprocal Rank Fusion, achieving 95% top-k accuracy.
- Designed agentic, plan-first workflows for SQL generation, validation, code execution, and answer synthesis to handle multi-step queries and follow-ups reliably.
- Deployed the system as an analyst-facing tool and supported release-readiness with testing, debugging, LangSmith tracing, and CI/Jenkins checks.

Laboratoire Interdisciplinaire des Sciences du Numérique (LISN) | Orsay, France

AI Research Intern — Speech Synthesis

February 2024 – August 2024

- Developed the first text-to-speech system for Walloon, an under-resourced minority language, using a VITS-based architecture in a low-resource setting.
 - Ran comparative experiments across grapheme- and phoneme-based inputs, including fine-tuning from French-pretrained models, to identify robust approaches for low-resource speech synthesis.
 - Evaluated the system with objective and human perceptual metrics, achieving a 4.22/5 MOS, and published the work as first author at SSW 2025 with models and demos released publicly.
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EDUCATION

Université Paris-Saclay | France

2023 – 2025

M.Sc. in Computer Science, Artificial Intelligence

Grade: 16.2/20

Universidad Industrial de Santander | Colombia

2017

B.Sc. in Electronic Engineering

PROJECTS & PUBLICATIONS

AskData: Analytics Assistant

2026

Built AskData, a full-stack analytics assistant for querying PostgreSQL data in plain English, combining FastAPI, Next.js, schema-aware retrieval, SQL validation, retry/repair handling, CI-backed tests, and benchmarked examples.

Speech Synthesis for Walloon, an Under-Resourced Minority Language

SSW 2025

First-author publication on the first Walloon text-to-speech system, using a VITS-based architecture and comparing grapheme, phoneme, and French-pretrained configurations with objective and perceptual evaluation.

SKILLS

Programming: Python, SQL, Bash, TypeScript.

AI / ML: PyTorch, scikit-learn, Pandas, NumPy, Matplotlib, Hugging Face Transformers.

LLM Systems: RAG, Agentic Workflows, LangChain, LangGraph, LangSmith, ChromaDB, BM25, Vector Databases, Prompt Engineering, OpenAI API, LLM Evaluation.

Tools: Git, Linux, Jupyter, Streamlit, FastAPI, Next.js, Pydantic, Docker, PostgreSQL, Oracle, Redis, CI, Slurm, Apify, Claude Code.

Languages: Spanish (Native), English (C1), French (B2), Portuguese (B2).